



Department of Computer Science and Information Technology

Campus Aware Intelligent AI Agent

Campus-Aware Intelligent AI Agent

User Manual

BY

Saiful Islam Shihab – 21732977
Sneh Manandhar – 21962370
Syed Mohammad Sadaat – 21808539
Samar Veer Singh – 21377817
Ifrat Bin Hasan Ruhan – 21873063

FOR:

2026-CSE3CAP (BU-1/OL-1) – Capstone Project
La Trobe University, Bundoora Campus

UNDER THE GUIDANCE OF:

Lecturer: Scott Mann
Project Owners: Di Wu & Phu Lai
Cisco – La Trobe Centre for AI & IoT

Version 1.0.0 | 2026

Copyright & Licensing

This document is produced as part of the 2026 Capstone Project (CSE3CAP) at La Trobe University by the Stack Over Five team.

Campus-Aware AI is developed for academic and research purposes under the Cisco – La Trobe Centre for AI & IoT. The software is intended for use within La Trobe University's Bundoora campus environment and is not intended for commercial distribution.

This program is distributed in the hope that it will be useful, but without warranty of any kind. Reproduction of this document or the software, in whole or in part, requires prior written permission from the project supervisors at La Trobe University.

Table of Contents

Copyright & Licensing	2
1.0 - Introduction	5
1.1 Audience Description	5
1.2 Applicability Statement	5
1.3 Purpose Statement	5
1.4 Document Usage Description	6
1.5 Conventions	6
1.6 System Requirements	7
2.0 - Introductory Kit	8
2.1 Campus AI Overview	8
2.2 Key Features	9
2.3 User Roles and Permissions	10
2.4 Getting Started	10
Option A – Continue as Guest	10
Option B – Create a student or Staff Account	10
Option C – Log in to an Existing Account	10
3.0 - Campus AI User Manual	11
3.1 Login	11
Screen Elements	12
Possible Actions	12
3.2 Create Account	12
Screen Elements	13
Possible Actions	13
3.3 Home / Chat	13
Left Sidebar	14
Main Chat Area	14
Possible Actions	14
3.4 Active Conversation	15
Conversation Layout	15
Supported Query Types	15
Response Quality	16
Possible Actions	16
3.5 User Menu	16
Menu Options	17
3.6 Profile	17
Profile Elements	17
Possible Actions	18
3.7 Settings	18

Settings Options	18
Possible Actions	19
3.8 Give Feedback	19
Form Details	20
Possible Actions	20
3.9 Sample Use Cases.....	21
Use Case 1 – New Student: Finding a Building.....	21
Use Case 2 – Registered Student: Academic Policy Query	21
Use Case 3 – Staff Member: Checking Campus Conditions	21
Use Case 4 – Guest User: Emergency Contact Lookup.....	22
Use Case 5 – Student: Finding Food and Facilities.....	22
3.10 Accessibility Features.....	22
4.0 - Installation Guide	24
5.0 - System Administrator Guide	25
5.1 System Architecture Overview.....	25
5.2 Intent Routing Architecture	25
5.3 Docker Setup.....	25
5.4 Backend Setup	26
5.5 Environment Variables	26
5.6 API Endpoints Reference	27
5.7 Database Maintenance.....	27
5.8 End-User Troubleshooting.....	27
5.9 Security Considerations.....	28
6.0 - Appendices.....	29
Appendix 1 – Table of Figures.....	29
Appendix 2 – Table of Tables	29
Appendix 3 – List of Abbreviations	29
7.0 - Bibliography.....	31
8.0 - Glossary	32
9.0 - Index.....	35

1.0 - Introduction

1.1 Audience Description

This document is intended for the following audience:

General Users (Students & Staff):

La Trobe University students and staff who wish to use the Campus AI chatbot to find campus information, navigate buildings, check live sensor data, or query university policies and academic information. These users require no prior technical knowledge to operate the system.

Guest Users:

Members of the public or new users who wish to explore the Campus AI system without creating an account. Guest users have limited functionality and their chat history is not saved.

System Administrators:

Technical staff or developers responsible for deploying, configuring, and maintaining the backend infrastructure. This audience should have familiarity with Docker, Python, and command-line tools.

1.2 Applicability Statement

The Campus AI application is accessible via the Expo Go mobile application on iOS and Android devices. The system is also accessible via a web browser during local development. The system is designed specifically for La Trobe University's Bundoora campus environment.

The system relies on:

- La Trobe University's campus IoT sensor infrastructure for live environmental data.
- Official La Trobe University PDF documentation as the RAG knowledge base.
- Google Gemini 2.5 Flash is the underlying large language model.
- PostgreSQL with pgvector for knowledge retrieval and InfluxDB for sensor telemetry.

The system supports both authenticated users (students and staff) who benefit from persistent chat history, and unauthenticated guest users with session-limited access.

1.3 Purpose Statement

The Campus-Aware Intelligent AI Agent was developed to address a common pain point for students and staff at La Trobe University: the difficulty of quickly finding accurate, up-to-date campus information across multiple sources. Whether navigating to a building, finding a quiet study space, checking live campus conditions, or understanding university policies, users previously had to search multiple websites and documents.

Campus AI consolidates all of this into a single, intelligent conversational interface. The major capabilities are:

- Campus navigation — directing users to buildings, rooms, facilities, and services across the Bundoora campus using official campus map data.
- Live IoT sensor data — querying real-time environmental readings such as temperature from campus-deployed IoT sensors via InfluxDB.
- RAG-based knowledge queries — answering questions drawn from 25+ official La Trobe University PDFs including policies, student guides, campus maps, and FAQs.
- Academic information — providing details on grades, census dates, special consideration, HELP fees, and other student services.
- Emergency information — surfacing critical campus contacts such as La Trobe Emergency and Campus Security.
- Persistent chat history — allowing logged-in users to revisit, rename, and manage previous conversations.

1.4 Document Usage Description

This User Manual is structured to serve different audiences at different points of need. The following sections are included:

1. Introduction – Overview of the system, its users, purpose, and document conventions.
2. Introductory Kit – System overview with architecture diagram, feature highlights, and getting started guide.
3. User Manual – Comprehensive screen-by-screen guide covering all features and user interactions.
4. Installation Guide – Steps required to set up and access the Campus AI application locally.
5. System Administrator Guide – Technical guide for backend deployment and system maintenance.
6. Appendices – Tables of figures, tables, and abbreviations.
7. Bibliography – Reference and source materials.
8. Glossary – Definitions of technical and domain-specific terms used in this document.
9. Index – Alphabetical reference index for quick lookup.

1.5 Conventions

The following formatting conventions are used throughout this document:

- Bold text highlights key terms, UI element names, button labels, and important notes.
- Numbered lists are used for sequential steps or ordered procedures.
- Bullet points are used for non-sequential lists of features or items.
- Screen names are written in Title Case to distinguish them from general text (e.g. Home / Chat screen).
- Code blocks appear in monospace font with a grey background (e.g. `docker compose up -d`).
- All figures are numbered and captioned below the image for easy reference.
- Tip and Note boxes appear in highlighted panels to draw attention to important information.

1.6 System Requirements

The following minimum requirements apply for end users accessing Campus AI:

Component	Requirement
Web Browser	Google Chrome 90+, Safari 14+, Firefox 88+, or Microsoft Edge 90+
Internet Connection	Required for all features (live IoT data requires stable connection)
Mobile OS (Expo Go)	iOS 13+ or Android 8.0+ with Expo Go installed
Screen Resolution	1024×768 minimum (recommended 1280×720 or higher)
JavaScript	Must be enabled in browser settings

Table 1. End User System Requirements

Note: No software installation is required for web browser access. Campus AI runs entirely within the browser without plugins or extensions.

2.0 - Introductory Kit

2.1 Campus AI Overview

Campus-Aware AI is an intelligent chatbot assistant developed at the Cisco – La Trobe Centre for AI & IoT by the Stack Over Five team as part of the 2026 Capstone Project at La Trobe University. The system is designed as a chatbot-first campus assistant system. All interactions are conducted through a natural language conversational interface, allowing users to ask questions the same way they would ask another person.

The system processes each user query through an intent routing orchestrator that determines the most appropriate data source and response strategy. This means the same chat interface handles navigation, live sensor data, university policy, and general campus knowledge seamlessly.

The system is built around three core architectural layers:

10. Frontend Interface Layer: A React Native Expo cross-platform application providing the chat interface through which users submit natural language queries and receive structured responses. Accessible via web browser or the Expo Go mobile application.
11. Logic / AI Layer: A FastAPI backend powered by Google Gemini 2.5 Flash with a Retrieval-Augmented Generation (RAG) pipeline. This layer handles intent routing, knowledge retrieval, sensor queries, and hallucination guardrails to ensure accurate responses.
12. Backend / Data Layer: PostgreSQL with pgvector for vector-based document storage (RAG), and InfluxDB for live IoT time-series sensor data. Firebase Authentication manages user identity and session security.

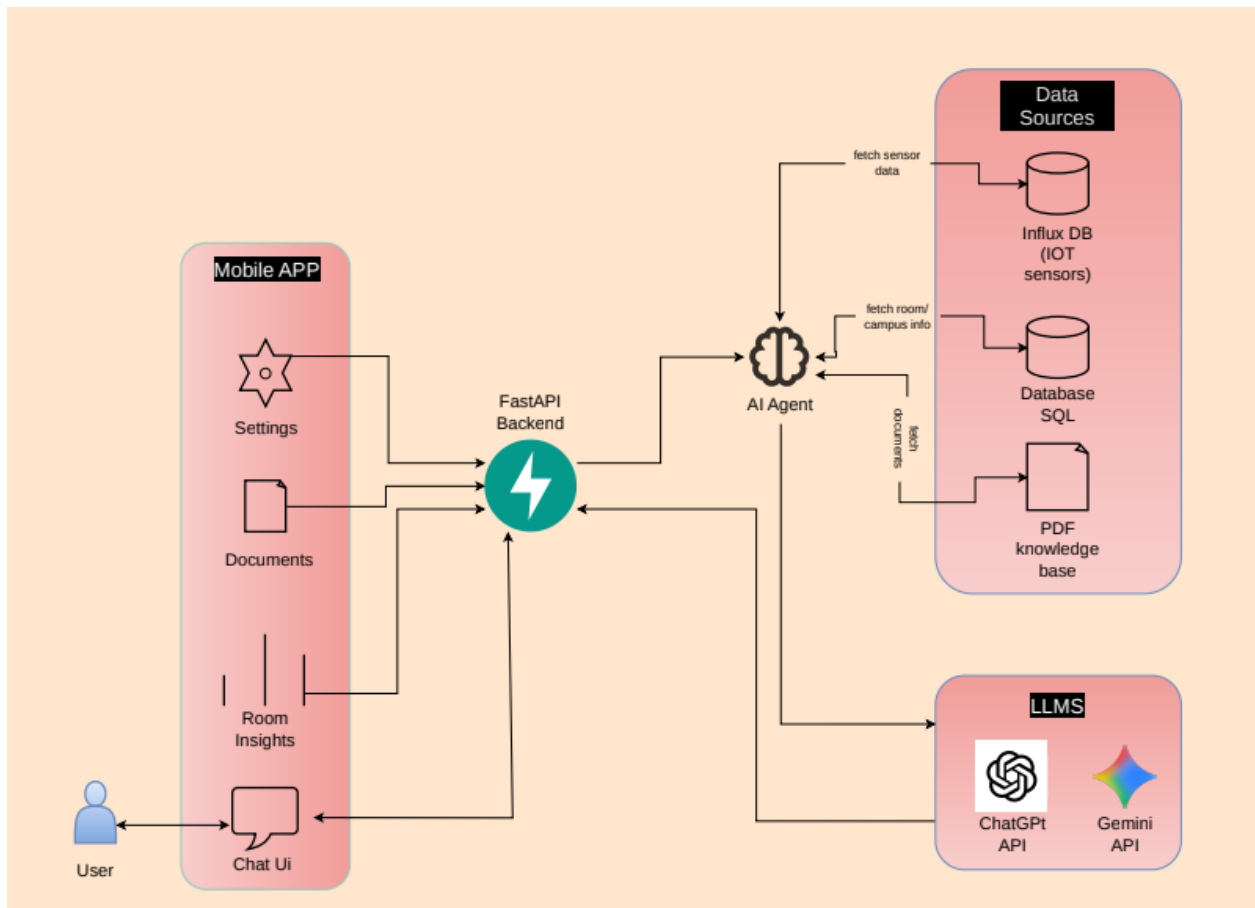


Figure 1. Campus-Aware AI System Architecture

2.2 Key Features

Feature	Description	Data Source
Campus Navigation	Natural language directions to buildings, rooms, and facilities	RAG – Campus Maps PDF
Live IoT Sensor Data	Real-time campus environmental readings (e.g. temperature)	InfluxDB (Live Sensors)
Policy & Academic Info	University policies, grades, census dates, special consideration	RAG – Policy PDFs
Emergency Contacts	Instant access to campus emergency and security numbers	RAG – Safety PDF
Facility Finder	Locate food, parking, toilets, study spaces, and services	RAG – Campus Guide
Chat History	Persistent, renameable conversation history for logged-in users	Firebase Firestore
Accessibility	Dark mode and large text toggle for improved readability	Frontend Settings
Feedback Collection	In-app link to usability testing feedback form	Google Forms

Table 2. Campus AI Key Features

2.3 User Roles and Permissions

Feature / Capability	Guest	Student	Staff
Ask campus navigation questions	✓	✓	✓
Query live IoT sensor data	✓	✓	✓
Query university policies & academic info	✓	✓	✓
Access emergency contacts	✓	✓	✓
Start a new chat session	✓	✓	✓
Persistent chat history across sessions	✗	✓	✓
Rename and manage chat sessions	✗	✓	✓
Profile page with recent chats	✗	✓	✓
Submit feedback via Google Form	✓	✓	✓
Toggle Dark Mode and Large Text	✓	✓	✓

Table 3. User Roles and Permissions

2.4 Getting Started

There are three ways to begin using Campus AI:

Option A – Continue as Guest

Select Continue as Guest on the login or create account screen. No registration required. You can ask questions immediately, but your chat history will not be saved when you close the app.

Option B – Create a student or Staff Account

Select Sign Up on the login screen. Fill in your name, choose Student or Staff, and enter your email and password. Optional fields include campus, courses, and year of study. Once registered, your chat history is saved and accessible from any device.

Option C – Log in to an Existing Account

If you have already registered, enter your email and password on the Login screen to access your account and previous chat history.

Tip: We recommend creating an account for the best experience. Chat history is only saved for registered users.

3.0 - Campus AI User Manual

This section provides a comprehensive guide to all screens and interactions available within the Campus AI application. All screens follow the red-and-white La Trobe University colour scheme. The application consists of 8 primary screens and a linked external feedback form.

#	Screen / Feature	Section
1	Login	3.1
2	Create Account	3.2
3	Home / Chat	3.3
4	Active Conversation	3.4
5	User Menu	3.5
6	Profile	3.6
7	Settings	3.7
8	Give Feedback	3.8

Table 4. Application Screens

3.1 Login

When Campus AI is launched, users are presented with a landing screen offering three options: Login, Sign Up, and Continue as Guest. The Login screen is displayed after tapping the Login option, and allows registered users to authenticate and access their saved chat history and profile.

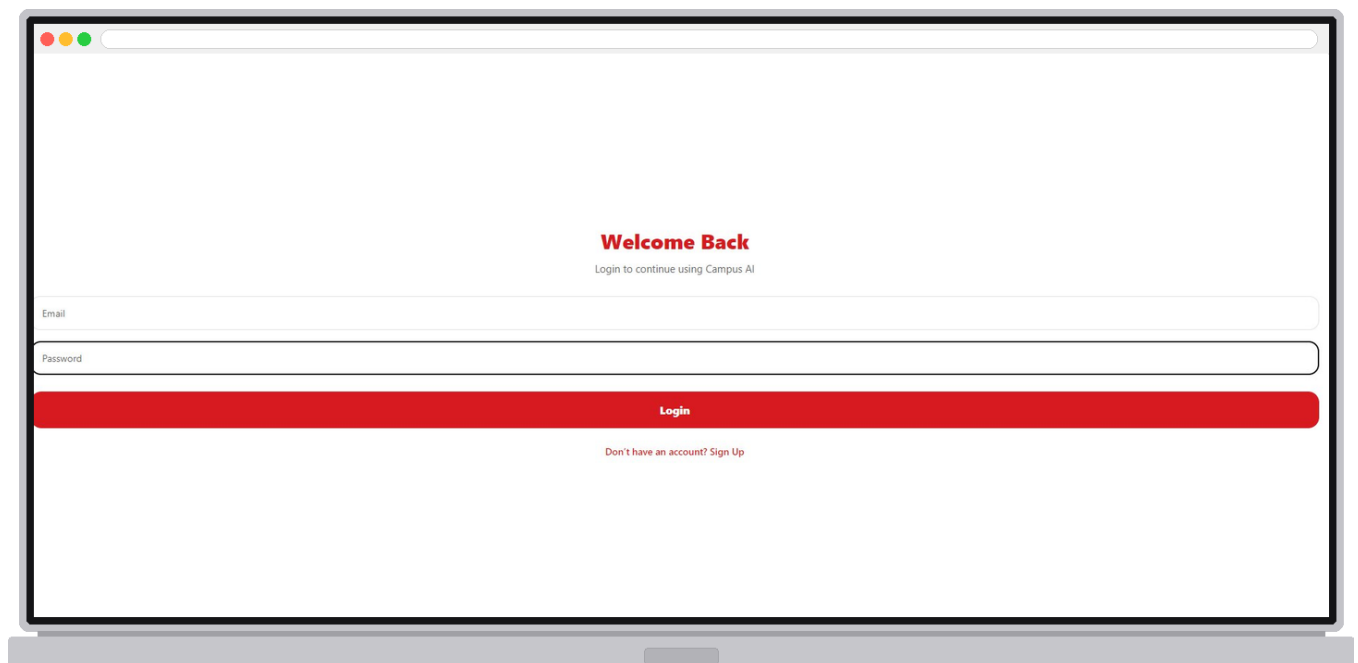


Figure 2. Campus AI Login Screen

Screen Elements

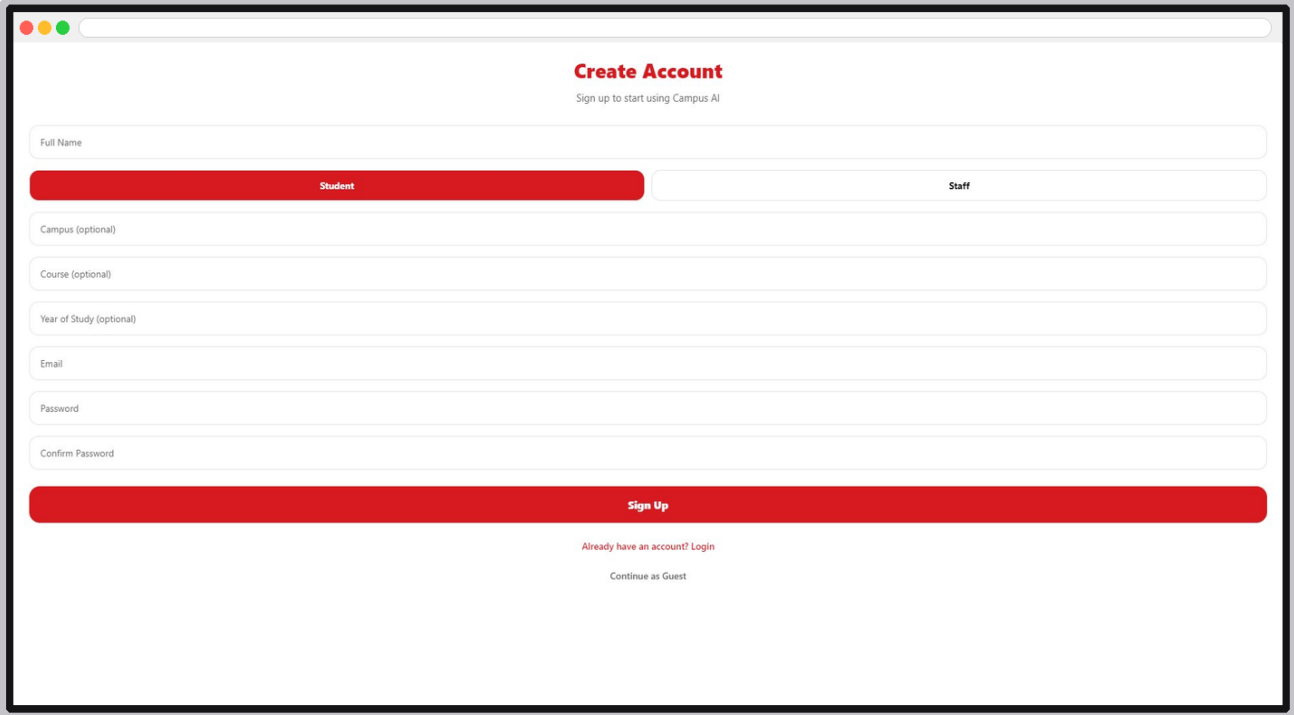
- Welcome Back heading in red.
- Subtitle: 'Login to continue using Campus AI'.
- Email text input field.
- Password text input field (masked).
- Red Login button spanning full width.
- 'Don't have an account? Sign Up' link in red below the button.

Possible Actions

13. Enter a valid registered email and password, then tap Login — authenticates the user and navigates to the Home / Chat screen (see Section 3.3).
14. Enter an invalid email or incorrect password — an error message is displayed; the user remains on the Login screen and may retry.
15. Leave email or password empty and tap Login — a validation message prompts the user to complete required fields.
16. Tap 'Don't have an account? Sign Up' navigates to the Create Account screen (see Section 3.2).

3.2 Create Account

The Create Account screen allows new users to register with Campus AI. Users select their role (Student or Staff) and provide their personal details to create a persistent account with saved chat history.



The screenshot displays the 'Create Account' screen for Campus AI. At the top, the title 'Create Account' is in red, with the subtitle 'Sign up to start using Campus AI' below it. The form includes a 'Full Name' input field, followed by two radio buttons for 'Student' (selected) and 'Staff'. Below these are four optional input fields: 'Campus (optional)', 'Course (optional)', 'Year of Study (optional)', and 'Email'. The 'Email' field is followed by 'Password' and 'Confirm Password' fields. A large red 'Sign Up' button spans the width of the form. At the bottom, there are two links: 'Already have an account? Login' and 'Continue as Guest'.

Figure 3. Campus AI Create Account Screen

Screen Elements

- 'Create Account' heading in red.
- Subtitle: 'Sign up to start using Campus AI'.
- Full Name text input (required).
- Student / Staff toggle selector — Student is selected by default (highlighted in red).
- Campus text input (optional).
- Course text input (optional — displayed for Student role).
- Year of Study text input (optional — displayed for Student role).
- Email text input (required).
- Password text input, masked (required).
- Confirm Password text input, masked (required).
- Red Sign-Up button spanning full width.
- 'Already have an account? Login' link.
- 'Continue as Guest' plain text link.

Possible Actions

17. Complete all required fields (Full Name, Email, Password, Confirm Password) and tap Sign Up — creates a new account and navigates to the Home / Chat screen (see Section 3.3).
18. Enter mismatched passwords — an error message is displayed indicating passwords do not match.
19. Enter an email address already registered — an error message is displayed.
20. Leave required fields empty — validation messages prompt completion of required fields.
21. Tap 'Already have an account? Login' navigates back to the Login screen (see Section 3.1).
22. Tap 'Continue as Guest' opens a guest session on the Home / Chat screen without creating an account.

Note: Optional fields (Campus, Course, Year of Study) help personalize your experience but are not required to complete registration.

3.3 Home / Chat

The Home / Chat screen is the primary interface of Campus AI. It is displayed immediately after a successful login or when accessing the application as a guest. This is where all conversations with the Campus AI agent begin.

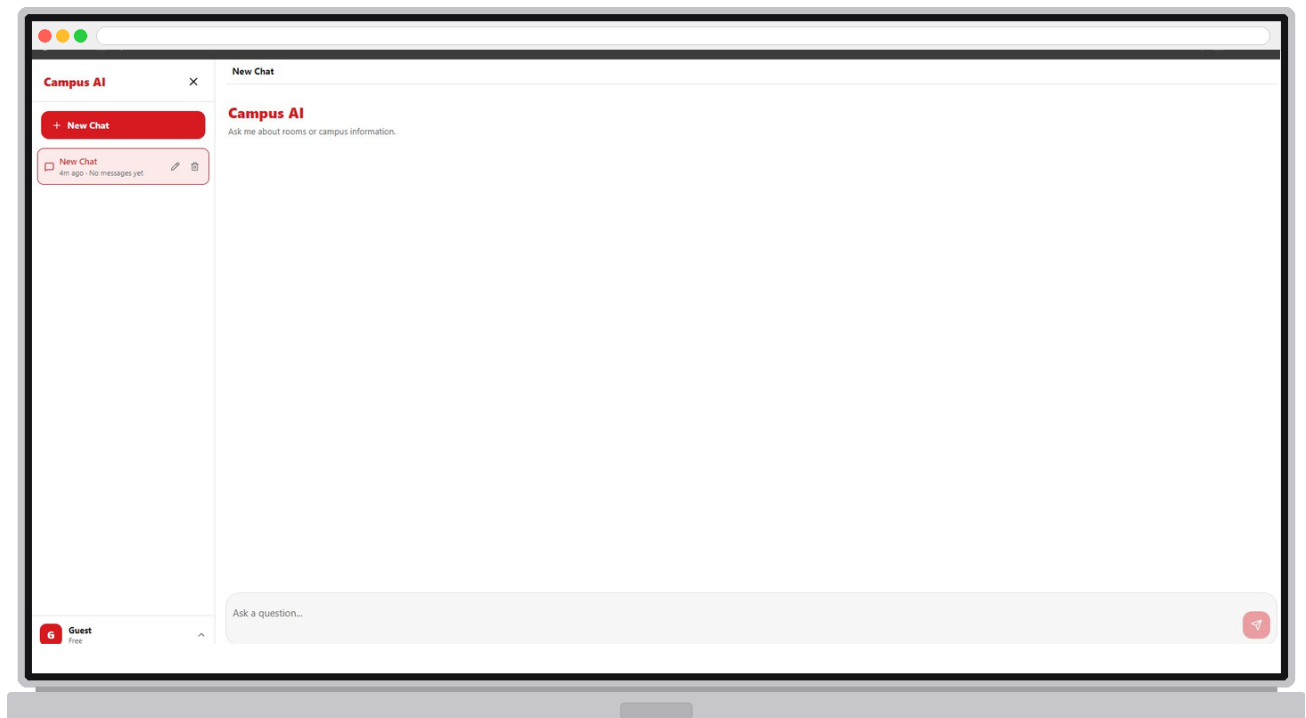


Figure 4. Campus AI Home / Chat Screen

Left Sidebar

- 'Campus AI' heading displayed at the top in red.
- Red '+ New Chat' button — starts a new conversation session.
- Chat history list showing all previous sessions for logged-in users, ordered by most recent. Each entry shows the session name and time elapsed since last activity.
- Rename icon (pencil) next to each chat — allows the user to rename the session.
- Delete icon (bin) next to each chat — permanently removes the session.
- User avatar and account type (Free / Guest) at the bottom left — clicking opens the User Menu (see Section 3.5).

Main Chat Area

- 'Campus AI' heading centered in the main area.
- Subtitle prompt: 'Ask me about rooms or campus information.'
- 'Ask a question...' text input bar at the bottom of the screen.
- Send button (arrow icon) to the right of the input bar.

Possible Actions

23. Tap '+ New Chat' starts a fresh conversation session.
24. Tap a previous chat from the sidebar — loads that conversation and displays its full history.
25. Tap the rename icon on a chat — opens an inline text field to rename the session.
26. Tap the delete icon on a chat — permanently removes that session after confirmation.
27. Type a question in the input bar and tap the send button — submits the query and navigates to the Active Conversation view (see Section 3.4).
28. Tap the user avatar at the bottom left — opens the User Menu popup (see Section 3.5).

Tip: Rename your chats with descriptive titles (e.g. 'Library Directions' or 'Census Date') to find them easily later from your profile.

3.4 Active Conversation

The Active Conversation view is displayed within the main chat area once a user has submitted one or more queries. The Campus AI agent processes each input through an intelligent intent routing system that determines whether to query the RAG knowledge base, live IoT sensors, historical sensor data, or the conversational LLM directly.

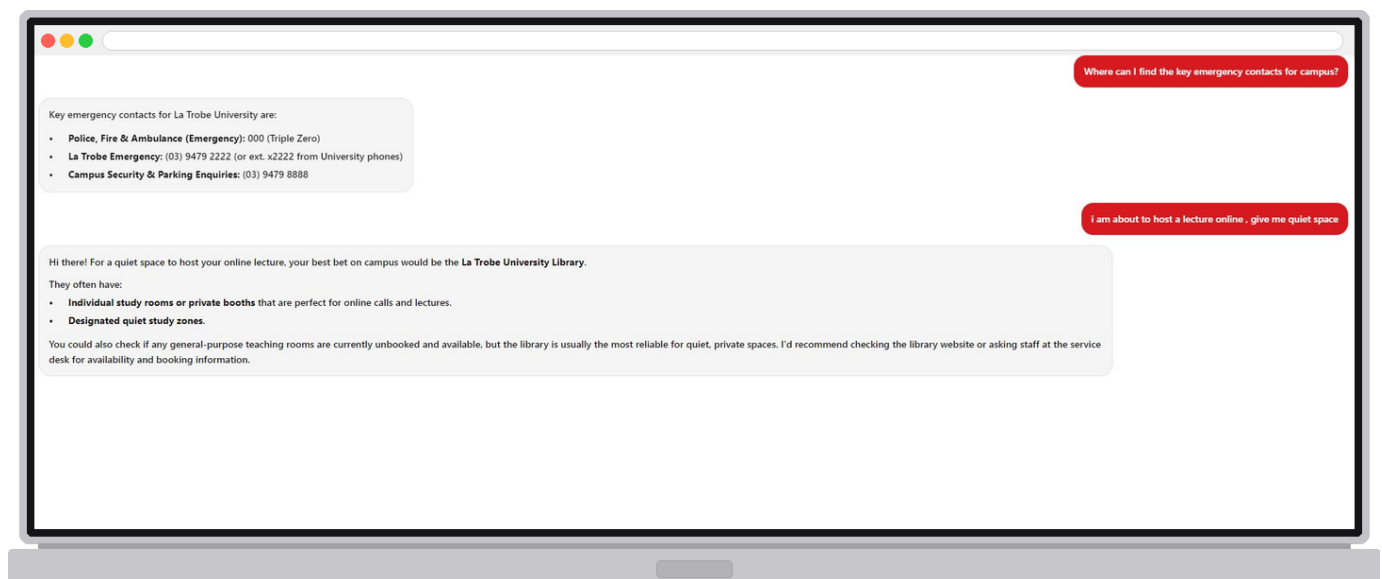


Figure 5. Campus AI Active Conversation Screen

Conversation Layout

- User messages appear as red rounded bubbles aligned to the right of the screen.
- AI responses appear as grey rounded cards aligned to the left of the screen.
- Responses may include formatted bullet points, bold text, and numbered lists for structured answers.
- Multiple conversation turns are displayed in chronological order, scrollable upward.

Supported Query Types

Query Type	Example Query	Routing	Data Source
Campus Navigation	How do I get to the library from Agora?	Navigation DB	PostgreSQL Navigation DB
Building / Room Lookup	Where is the La Trobe Medical Centre?	Navigation DB	PostgreSQL Navigation DB
Facility Finder	Where can I grab food or coffee on campus?	RAG	Campus Guide PDF
Emergency Contacts	What are the key emergency contacts?	RAG	Safety Guide PDF

Academic Policy	How are grades calculated at La Trobe?	RAG	Policy PDF
Student Services	How do I apply for special consideration?	RAG	Student Guide PDF
Census Date	What is census date and why does it matter?	RAG	Student Guide PDF
Live IoT Sensor	What is the current temperature on campus?	Sensor	InfluxDB Live
Historical Sensor	What was the average temperature last week?	Text-to-Flux	InfluxDB Historical
General Campus Info	What events are on campus this week?	Conversational	Gemini LLM

Table 5. Supported Query Types and Routing

Response Quality

Campus AI is designed with hallucination guardrails — if the system cannot confidently answer a query from the available knowledge base or sensor data, it will indicate uncertainty and recommend the user visit latrobe.edu.au or contact La Trobe student services directly.

Possible Actions

29. Type and submit another question in the input bar — the conversation continues in the same session.
30. Tap '+ New Chat' from the sidebar — starts a fresh conversation without ending the current one.
31. Scroll up — views earlier messages in the conversation history.
32. Tap a previous chat from the sidebar — navigates to a different conversation session.

Tip: You can ask follow-up questions naturally. For example, after asking 'Where is the library?' you can follow up with 'How do I get there from Building B?'

3.5 User Menu

The User Menu is a compact popup panel that appears when the user taps their avatar or username at the bottom-left corner of the screen. It provides quick access to account management and application options.

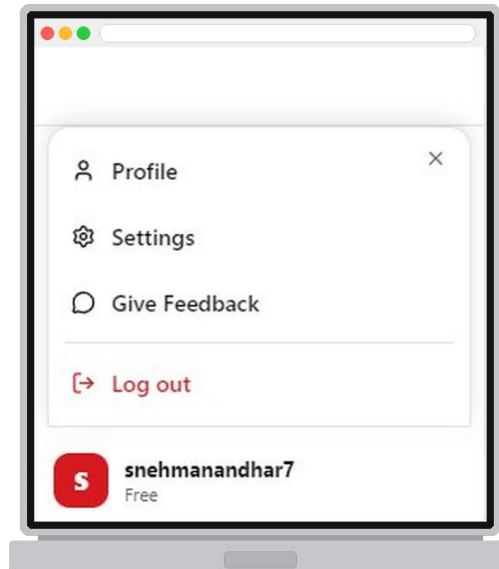


Figure 6. Campus AI User Menu

Menu Options

33. Profile — opens the Profile panel showing account details and recent chat history (see Section 3.6).
34. Settings — opens the Settings panel for appearance customization (see Section 3.7).
35. Give Feedback — opens the Campus AI Usability Testing Google Form in the browser (see Section 3.8).
36. Log Out — ends the current authenticated session and returns the user to the Login screen (see Section 3.1). Guest sessions are cleared.

Note: The User Menu is accessible from any screen within the application by tapping the avatar at the bottom left of the sidebar.

3.6 Profile

The Profile panel displays information about the currently authenticated user, including their account type and a list of recent chat sessions. It is accessible from the User Menu (see Section 3.5).



Figure 7. Campus AI Profile Screen

Profile Elements

- User avatar displaying the first letter of the username on a red background.
- Username and account type badge (e.g. Guest, Free).
- 'Browsing as guest' label for unauthenticated users.
- Recent Chats section listing all saved conversations with session name and date.
- Delete (bin) icon next to each chat entry for permanent removal.
- Footer: 'Campus-Aware AI · Cisco – La Trobe Centre for AI & IoT · v1.0.0'.

Possible Actions

37. Tap a recent chat entry — closes the Profile panel and navigates to that conversation.
38. Tap the delete icon next to a chat — permanently removes that session.
39. Tap the X (close) button — closes the Profile panel and returns to the Home / Chat screen.

Note: Chat history is only available for registered users. Guest users see an empty Recent Chats list and the 'Browsing as guest' indicator.

3.7 Settings

The Settings panel allows users to customize the visual appearance of the Campus AI application. It is accessible from the User Menu (see Section 3.5).

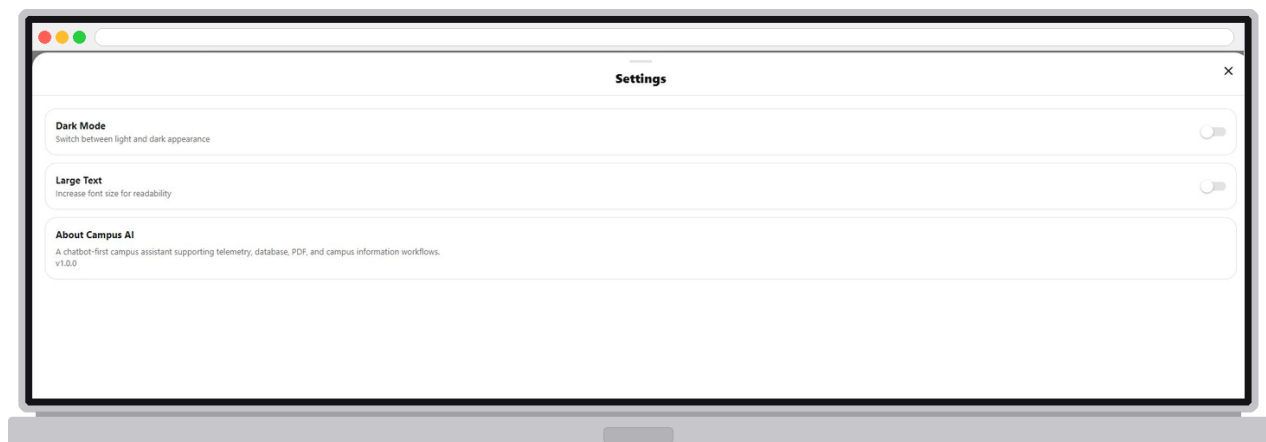


Figure 8. Campus AI Settings Screen

Settings Options

Setting	Description	Default
Dark Mode	Switches the entire application interface between light and dark color themes. Dark mode reduces eye strain in low-light environments.	Off (Light Mode)
Large Text	Increases the font size across the application for improved readability. Beneficial for users with vision impairments.	Off (Standard)
About Campus AI	Displays a brief description of the application: 'A chatbot-first campus assistant supporting telemetry, database, PDF, and campus information workflows.'	v1.0.0

	database, PDF, and campus information workflows.' Shows the current version number.	
--	---	--

Table 6. Settings Options

Possible Actions

- 40. Toggle Dark Mode on — immediately applies dark color theme to the entire application.
- 41. Toggle Dark Mode off — reverts to the default light color theme.
- 42. Toggle Large Text on — immediately increases font size across all screens.
- 43. Toggle Large Text off — reverts to standard font size.
- 44. Tap the X (close) button — closes the Settings panel and returns to the Home / Chat screen.

Tip: Settings preferences are applied immediately and persist for the duration of the session. Registered users may find their preferences persist across sessions.

3.8 Give Feedback

The Give Feedback option in the User Menu opens an external Google Form to collect usability feedback from participants. This feature supports the ongoing quality assurance and usability evaluation of the Campus AI system conducted by the Stack Over Five team.

Campus-Aware Intelligent AI Agent Project
Usability Testing

snehanandhar7@gmail.com [Switch account](#)
Not shared

PARTICIPANT DETAIL
Tell us a little about yourself so we can make sense of the responses.

What's your full name?
Your answer

Are you a staff or a student at the LaTrobe?
☐ Student
☐ Staff

How familiar are you with AI chatbots or virtual assistants before this session?
☐ Not Familiar
☐ Slightly Familiar
☐ Moderately Familiar

[Next](#) [Clear form](#)

Never submit passwords through Google Forms.
This content is neither created nor endorsed by Google. [Contact form owner](#) [Terms of Service](#) [Privacy Policy](#)
Does this form look suspicious? [Report](#)

Google Forms

Figure 9. Campus AI Usability Testing Feedback Form

Form Details

The form is titled 'Campus-Aware Intelligent AI Agent Project – Usability Testing' and collects the following participant information:

- Full Name.
- Role at La Trobe University: Student or Staff.
- Prior familiarity with AI chatbots or virtual assistants: Not Familiar / Slightly Familiar / Moderately Familiar.

Possible Actions

45. Complete the participant details and tap Next — proceeds to the task-based questions section of the form.
46. Tap Clear form — clears all entered responses.
47. Complete all sections and submit the form — responses are recorded in the Stack Over Five team's usability data collection system.

Note: Feedback is collected anonymously for research purposes and is used solely to improve the Campus AI system. No personal data beyond name and role is collected.

3.9 Sample Use Cases

This section presents five realistic step-by-step walkthroughs demonstrating how different users interact with Campus AI to accomplish common tasks. These scenarios illustrate the breadth of the system's capabilities.

Use Case 1 – New Student: Finding a Building

Step	User Action	System Response
1	Opens Campus AI in the browser and taps 'Continue as Guest'.	Navigate to the Home / Chat screen with guest session.
2	Types: 'Where is the Engineering building on campus?'	AI responds with building location, including the building code and nearby reference points.
3	Follows up: 'How do I get there from the Agora?'	AI provides step-by-step walking directions from the Agora to the Engineering building.
4	Asks: 'Is there parking nearby?'	AI provides the nearest car park codes and their locations related to the building.

Use Case 2 – Registered Student: Academic Policy Query

Step	User Action	System Response
1	Logs in with student credentials and taps '+ New Chat'.	Home / Chat screen loads with the student's previous chat history in the sidebar.
2	Types: 'What is the census date and what happens if I miss it?'	AI explains census date, deadlines, financial implications, and HELP fee obligations.
3	Asks: 'How do I apply for special consideration for an assignment?'	AI provides the four-step process for applying for special consideration with relevant deadlines.
4	Renames the chat 'Academic Policies' using the pencil icon.	Chat is saved with the new name and appears in the sidebar and Profile.

Use Case 3 – Staff Member: Checking Campus Conditions

Step	User Action	System Response
1	Log in as Staff and starts a new chat.	Home / Chat screen loads.
2	Types: 'What is the current temperature on campus?'	AI queries InfluxDB live sensors and returns: 'The current temperature on campus is 28.8°C.'
3	Asks: 'Where is a quiet space I can host an online meeting?'	AI recommends the La Trobe University Library individual study rooms and private booths.

4	Opens Settings and enables Dark Mode for a more comfortable viewing experience.	Application immediately switches to dark color theme.
---	---	---

Use Case 4 – Guest User: Emergency Contact Lookup

Step	User Action	System Response
1	Opens Campus AI and continues as guest.	Home / Chat screen loads.
2	Types: 'Where can I find the key emergency contacts for campus?'	AI returns: Police/Fire/Ambulance (000), La Trobe Emergency (03 9479 2222), Campus Security (03 9479 8888).
3	Asks: 'Where is the nearest medical Center on campus?'	AI responds: 'The La Trobe Uni Medical Centre is located at HS1 Level 1.'

Use Case 5 – Student: Finding Food and Facilities

Step	User Action	System Response
1	Logs in and starts a new chat.	Home / Chat screen loads.
2	Types: 'Where can I grab food or coffee on campus?'	AI responds: 'You can grab food and beverages at the Food Court, located in Agora West and Agora East.'
3	Ask: 'Where are the nearest toilets from the Agora?'	AI provides locations: Agora building (A), Agora Theatre (AT) on levels 1–2 (incl. Gender Neutral), and accessible toilet near Learning Commons (DSA-02).
4	Submit feedback via Give Feedback after the session.	Google Form opens in the browser for usability feedback submission.

Table 7. Sample Use Case Walkthroughs

3.10 Accessibility Features

Campus AI is designed with accessibility in mind. The following features are available to all users regardless of account type:

Feature	How to Access	Benefit
Dark Mode	User Menu → Settings → Toggle Dark Mode	Reduces eye strain in low-light environments. Recommended for night-time use.
Large Text	User Menu → Settings → Toggle Large Text	Increases font size across all screens. Beneficial for users with visual impairments.
Guest Access	Login or Create Account screens Continue as Guest	Allows users without accounts to access all core query features immediately.

Keyboard / Touch	All interactions via touch or keyboard	Fully operable without a mouse on desktop and touch devices.
Structured Responses	Active Conversation screen	AI formats responses with headings, bullet points, and numbered steps for clarity.

Table 8. Accessibility Features

Note: If you require additional accessibility support, please contact La Trobe University's Accessibility and Disability Resource Centre (ADRC) for assistance.

4.0 - Installation Guide

The Campus-Aware Intelligent AI Agent has not been publicly deployed at this stage, as per the project supervisor's instruction. The System Administrator Guide (Section 5.0) provides detailed instructions for setting up the system locally for development and demonstration purposes.

The backend and mobile application are fully self-contained and would need to be deployed to an online hosting service or run locally via Docker Compose when required. The mobile application requires a running backend server, which connects to PostgreSQL (with pgvector) and InfluxDB via Docker Compose.

Required libraries and software for the backend:

API Server:

- fastapi — web framework for the REST API
- uvicorn — ASGI web server to serve the FastAPI application
- python-dotenv — environment variable management
- pydantic — data validation and settings management

AI and Knowledge Retrieval:

- google-genai — Google Gemini 2.5 Flash large language model integration
- sentence-transformers (all-MiniLM-L6-v2) text embeddings for RAG
- PyMuPDF — PDF document parsing for knowledge base ingestion
- psycopg2 — PostgreSQL database connector
- influxdb-client — InfluxDB time-series database connector

Infrastructure (via Docker Compose):

- PostgreSQL 16 + pgvector — vector database for RAG document storage
- InfluxDB 2.7 — time-series database for live IoT sensor data

Mobile Frontend:

- Node.js 20 LTS — JavaScript runtime for the Expo project
- pnpm — fast Node.js package manager
- Expo Go — mobile application (Apple App Store / Google Play Store) to run the app on a physical device

End users do not need to install any software to access Campus AI through a web browser once the system is deployed. Users who prefer native mobile access may install Expo Go on their smartphone and scan the QR code provided by the system administrator.

For full setup and deployment instructions, refer to Section 5.0 – System Administrator Guide.

5.0 - System Administrator Guide

This section is intended for system administrators and developers responsible for deploying and maintaining the Campus AI backend infrastructure.

5.1 System Architecture Overview

Layer	Component	Technology	Port
Frontend	Mobile / Web Chat Interface	React Native Expo	–
API Layer	REST API Server	FastAPI / uvicorn	8000
AI Model	Intent Routing + LLM	Google Gemini 2.5 Flash	–
Knowledge Base	Vector Database (RAG)	PostgreSQL + pgvector (Docker)	5432
Telemetry	IoT Sensor Data	InfluxDB (Docker)	18086
Authentication	User Auth & Sessions	Firebase Authentication	–

Table 9. System Architecture Components

5.2 Intent Routing Architecture

Every user query is classified by the intent routing orchestrator before being sent to a data source. The routing logic follows this priority:

48. If the query relates to live sensor data (e.g. current temperature) → route to InfluxDB live sensor endpoint.
49. If the query relates to historical sensor data (e.g. average temperature last week) → convert to Flux query via Text-to-Flux pipeline and query InfluxDB.
50. If the query intent is navigation_where, navigation_directions, or navigation_nearest → route directly to the PostgreSQL Navigation Database (bypasses RAG entirely).
51. If the query intent is exact_current_info → apply trusted-source guardrail, extract verified URLs from RAG context, and return only safe referenced links.
 - If the query intent is rag_specific → high-confidence RAG returns answer directly from PostgreSQL/pgvector knowledge base.
 - If the query intent is general_conceptual → use hybrid RAG + LLM (Gemini) combining retrieved chunks with general knowledge.
 - If the query is conversational or general → send directly to Google Gemini with hallucination guardrails active.

5.3 Docker Setup

All backend database services run as Docker containers managed by Docker Compose.

52. Clone the project repository:

```
git clone https://github.com/Saiful-Codes/campus-aware-ai-agent.git
cd campus-aware-ai-agent
```

53. Start all database services:

```
docker compose up -d
```

54. Verify containers are running:

```
docker ps
```

This starts PostgreSQL on port 5432 and InfluxDB on port 18086.

5.4 Backend Setup

55. Navigate to the backend folder and create a virtual environment:

```
cd backend
python -m venv .venv
.venv\Scripts\Activate.ps1    # PowerShell
```

56. Install dependencies:

```
pip install -r requirements.txt
```

57. Copy and configure the environment file:

```
copy ..\.env.example .env
```

58. Start the API server:

```
uvicorn app.main:app --reload --host 0.0.0.0 --port 8000
```

5.5 Environment Variables

Variable	Description	Example Value
APP_ENV	Application environment mode (controls strict validation on startup)	development
APP_HOST	Host address for the API server	0.0.0.0
APP_PORT	Port number for the API server	8000
GEMINI_API_KEY	Google Gemini API key (required)	your_gemini_api_key
POSTGRES_HOST	PostgreSQL host address	localhost
POSTGRES_PORT	PostgreSQL port	5432
POSTGRES_DB	Database name	campus_ai
POSTGRES_USER	Database username	postgres
POSTGRES_PASSWORD	Database password	postgres
INFLUXDB_URL	InfluxDB connection URL	http://localhost:18086
INFLUXDB_TOKEN	InfluxDB authentication token	campus-ai-token-123
INFLUXDB_ORG	InfluxDB organisation name	campus-ai
INFLUXDB_BUCKET	InfluxDB bucket name	sensor_data

SENSOR_SYNC_ENABLED	Enable/disable IoT sensor sync loop	true
SENSOR_SYNC_INTERVAL_SECONDS	Sensor polling interval in seconds	300

Table 10. Backend Environment Variables

5.6 API Endpoints Reference

Method	Endpoint	Description
GET	/health	Returns backend health status and environment info
POST	/chat	Main intent-orchestrated chat endpoint — routes to RAG, sensor, or LLM
POST	/rag/chat	Direct RAG knowledge base chat — bypasses intent routing
GET	/sensor/latest	Returns the latest live IoT sensor reading from InfluxDB

Table 11. Backend API Endpoints

Interactive API documentation (Swagger UI) is available at <http://127.0.0.1:8000/docs> when the backend is running.

5.7 Database Maintenance

PostgreSQL / pgvector (RAG Knowledge Base):

The RAG knowledge base is populated from official La Trobe University PDF documents using the PyMuPDF parser and the all-MiniLM-L6-v2 sentence transformer for embeddings. To update the knowledge base with new documents, add PDFs to the backend/ragData/ folder and re-run the ingestion pipeline script.

InfluxDB (IoT Telemetry):

IoT sensor data is ingested in real time from campus-deployed sensors. The SENSOR_SYNC_ENABLED and SENSOR_SYNC_INTERVAL_SECONDS environment variables control the ingestion loop. Set SENSOR_SYNC_ENABLED=false during testing to prevent sensor loop noise.

5.8 End-User Troubleshooting

Issue	Likely Cause	Resolution
'I can't log in' – error message shown	Incorrect email or password	Check credentials and retry. Ensure Caps Lock is off. Contact admin if issue persists.

'Network request failed' on mobile	Phone cannot reach backend (Expo Go)	Ensure phone and laptop are on the same Wi-Fi. Use laptop's IPv4 address, not localhost.
Chat history not showing	User is in guest mode	Create a registered account to save chat history across sessions.
AI response seems inaccurate	Query outside knowledge base scope	Rephrase the question or visit latrobe.edu.au for official information.
Apps are slow to respond	Network latency or server load	Wait a few seconds and retry. Check internet connection.
Settings not saving	Session-specific settings	Re-apply settings after refresh. Registered user preferences may persist.
Temperature query returns error	InfluxDB container not running	Admin: run 'docker compose up -d' to restart database services.

Table 12. End-User Troubleshooting Guide

5.9 Security Considerations

- User passwords are not stored in plain text. Firebase Authentication handles all credential management securely.
- API keys (GEMINI_API_KEY, INFLUXDB_TOKEN) must be stored in .env files and must never be committed to version control.
- The .gitignore file is configured to exclude backend/.env, mobile/.env, .venv, and node_modules.
- The InfluxDB token (campus-ai-token-123) is the default development value and must be changed before any production deployment.
- All API endpoints should be placed behind an HTTPS reverse proxy (e.g. Nginx) in any production environment.
- Hallucination guardrails are active on all LLM responses to reduce the risk of the system providing dangerous or misleading information.

6.0 - Appendices

Appendix 1 – Table of Figures

Figure No.	Figure Name	Section
Figure 1	Campus-Aware AI System Architecture	2.1
Figure 2	Campus AI Login Screen	3.1
Figure 3	Campus AI Create Account Screen	3.2
Figure 4	Campus AI Home / Chat Screen	3.3
Figure 5	Campus AI Active Conversation Screen	3.4
Figure 6	Campus AI User Menu	3.5
Figure 7	Campus AI Profile Screen	3.6
Figure 8	Campus AI Settings Screen	3.7
Figure 9	Campus AI Usability Testing Feedback Form	3.8

Appendix 2 – Table of Tables

Table No.	Table Name	Section
Table 1	End User System Requirements	1.6
Table 2	Campus AI Key Features	2.2
Table 3	User Roles and Permissions	2.3
Table 4	Application Screens	3.0
Table 5	Supported Query Types and Routing	3.4
Table 6	Settings Options	3.7
Table 7	Sample Use Case Walkthroughs	3.9
Table 8	Accessibility Features	3.10
Table 9	System Architecture Components	5.1
Table 10	Backend Environment Variables	5.5
Table 11	Backend API Endpoints	5.6
Table 12	End-User Troubleshooting Guide	5.8

Appendix 3 – List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface

DB	Database
Docker	Docker containerisation platform
Expo	Expo Go development framework for React Native
FastAPI	Fast Application Programming Interface (Python framework)
IoT	Internet of Things
LLM	Large Language Model
PDF	Portable Document Format
pgvector	PostgreSQL vector similarity extension
pnpm	Performant Node Package Manager
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
YAML	YAML Ain't Markup Language

7.0 - Bibliography

2026-CSE3CAP (BU-1/OL-1) Capstone Project Handbook. La Trobe University, 2026.

Cisco – La Trobe Centre for AI & IoT. (2026). Project Brief: Campus-Aware Intelligent AI Agent. La Trobe University.

La Trobe University. (2026). Official Documentation: Campus Maps, Student Guides, Policies, and FAQs. Retrieved from <https://latrobe.edu.au>

Google. (2026). Gemini API Documentation. Retrieved from <https://ai.google.dev>

Expo. (2026). Expo Documentation. Retrieved from <https://docs.expo.dev>

Tiangolo, S. (2024). FastAPI Documentation. Retrieved from <https://fastapi.tiangolo.com>

pgvector. (2024). PostgreSQL Extension for Vector Similarity Search. Retrieved from <https://github.com/pgvector/pgvector>

InfluxData. (2026). InfluxDB 2.7 Documentation. Retrieved from <https://docs.influxdata.com>

Google Firebase. (2026). Firebase Authentication Documentation. Retrieved from <https://firebase.google.com/docs/auth>

Docker Inc. (2026). Docker Documentation. Retrieved from <https://docs.docker.com>

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. Proceedings of EMNLP 2019. Retrieved from <https://www.sbert.net>

Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale Similarity Search with GPUs. IEEE Transactions on Big Data.

8.0 - Glossary

API (Application Programming Interface)

A set of protocols and tools that allow software applications to communicate with each other. Campus AI uses a FastAPI REST API to connect the frontend and backend.

Chatbot

A software application that simulates human conversation through text. Campus AI uses an LLM-powered chatbot to respond to campus-related queries.

Docker

A platform for running applications in isolated containers. Campus AI uses Docker Compose to run PostgreSQL and InfluxDB locally.

Docker Compose

A tool for defining and running multi-container Docker applications using a YAML configuration file. Used in Campus AI to start all backend services with a single command.

Embeddings

Numerical vector representations of text that capture semantic meaning. Used in RAG to find document chunks most relevant to a user's query.

Expo Go

A mobile application that allows developers and users to run React Native Expo projects on physical iOS and Android devices by scanning a QR code.

FastAPI

A modern, high-performance Python web framework for building APIs. The Campus AI backend is built with FastAPI and served by uvicorn.

Firebase Authentication

Google's identity platform used by Campus AI to manage user registration, login, and session security on the mobile application.

Flux

A functional query language used by InfluxDB for querying and analysing time-series data. Campus AI uses Text-to-Flux to convert natural language sensor queries into Flux queries.

Gemini (Google Gemini 2.5 Flash)

Google's large language model used by Campus AI to generate natural language responses. Selected for its speed, accuracy, and cost-effectiveness.

Guest User

A user who accesses Campus AI without creating an account. Guest sessions are temporary and chat history is not saved.

Hallucination Guardrails

Safety mechanisms within the AI pipeline that detect and suppress responses that the LLM cannot confidently support from the knowledge base or sensor data.

InfluxDB

A time-series database optimised for storing and querying IoT sensor data. Campus AI uses InfluxDB to retrieve live and historical campus environmental readings.

Intent Routing

The process of classifying a user's query and directing it to the appropriate data source or processing pipeline — live sensor, historical sensor, RAG, or conversational LLM.

IoT (Internet of Things)

A network of physical devices with sensors and internet connectivity. Campus AI queries IoT sensors deployed across La Trobe University's Bundoora campus.

LLM (Large Language Model)

An AI model trained on large text datasets capable of understanding and generating human language. Campus AI uses Google Gemini as its LLM.

pgvector

A PostgreSQL extension enabling vector similarity search. Used in Campus AI to store document embeddings and retrieve the most relevant knowledge base chunks.

pnpm

A fast and disk-space-efficient Node.js package manager. Used to manage the Campus AI frontend (Expo) dependencies.

RAG (Retrieval-Augmented Generation)

A technique that combines a language model with a document retrieval system. When a user asks a question, relevant document chunks are retrieved and provided to the LLM to generate an accurate, grounded response.

React Native Expo

A framework for building cross-platform mobile and web applications using React. The Campus AI frontend is built with React Native Expo.

Sentence Transformer

A neural network model that converts sentences into dense vector embeddings. Campus AI uses the all-MiniLM-L6-v2 model for generating text embeddings in the RAG pipeline.

Telemetry

The automated collection and transmission of data from remote sensors. In Campus AI, telemetry refers to live campus environmental sensor data.

uvicorn

A lightning-fast ASGI web server for Python. Used to serve the FastAPI backend in the Campus AI system.

Vector Database

A database optimised for storing and searching high-dimensional vector embeddings. Campus AI uses PostgreSQL with the pgvector extension as its vector database.

9.0 - Index

The following is an alphabetical index of key terms, features, and screens referenced in this document.

A

Accessibility – 3.10, 8.0

API – 5.6, 8.0

Architecture – 2.1, 5.1

C

Campus AI – 1.3, 2.1

Change Log – Change Log page

Chat History – 1.3, 3.3, 3.6

Chatbot – 1.3, 8.0

Continue as Guest – 3.1, 3.2, 2.4

Create Account – 3.2

D

Dark Mode – 3.7, 3.10

Docker – 4.0, 5.3, 8.0

Docker Compose – 5.3, 8.0

E

Embeddings – 8.0

Emergency Contacts – 1.3, 3.9

Environment Variables – 5.5

Expo Go – 4.0, 8.0

F

FastAPI – 5.1, 8.0

Feedback – 3.5, 3.8

Firebase Authentication – 5.1, 8.0

Flux – 5.2, 8.0

G

Gemini – 2.1, 5.1, 8.0

Give Feedback – 3.5, 3.8

Guest User – 2.3, 2.4, 3.1, 3.2, 8.0

H

Hallucination Guardrails – 3.4, 5.9, 8.0

Home / Chat – 3.3

I

InfluxDB – 5.1, 5.5, 5.7, 8.0

Installation – 4.0

Intent Routing – 5.2, 8.0

IoT – 1.3, 2.1, 3.4, 8.0

K

Key Features – 2.2

L

La Trobe University – Cover, 1.2, 1.3

Large Text – 3.7, 3.10

LLM – 2.1, 5.2, 8.0

Login – 3.1

Log Out – 3.5

N

Navigation – 1.3, 2.2, 3.4, 3.9

New Chat – 3.3

P

Page Numbers – Footer

pgvector – 5.1, 5.7, 8.0

pnpm – 4.0, 8.0

Profile – 3.5, 3.6

R

RAG – 1.3, 2.1, 3.4, 5.2, 8.0

React Native Expo – 2.1, 5.1, 8.0

S

Sample Use Cases – 3.9

Security – 5.9

Sentence Transformer – 5.7, 8.0

Settings – 3.5, 3.7

Sign Up – 3.2

Stack Over Five – Cover, Statement of Effort

Statement of Effort – Statement of Effort page

Student – 1.1, 2.3, 3.2

System Requirements – 1.6

T

Telemetry – 2.1, 3.4, 5.7, 8.0

Temperature – 3.4, 3.9

Troubleshooting – 5.8

U

User Manual – 3.0

User Menu – 3.5

User Roles – 2.3

uvicorn – 5.1, 8.0

V

Vector Database – 5.1, 8.0

Version – Cover, 3.7